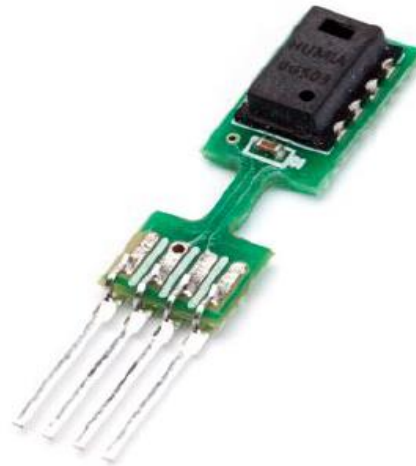


HCSD Series

Humidity & Temperature Solution

Features

- ◆ Fully Calibrated & Temperature Compensated
- ◆ Digital (I²C) Output
- ◆ Precision & Accuracy ($\pm 2\%$ RH, $\pm 0.3^{\circ}\text{C}$, 14bit)
- ◆ Free Operating Voltage (min 2.3V to max 5.5V)
- ◆ Low Current Consumption
- ◆ SIP (Single-In-line-Package)
- ◆ Reliable in Harsh Environment



Product Summary

HCSD Series is a Single In-Line Package type of **HCSD** with ready installed V-Core capacitor for easy and convenient application.

HCSD offers the most advanced and cost effective humidity and temperature sensing solution for virtually any type of applications.

Capacitive polymer sensor chip developed and fabricated in-house and CMOS integrated circuit with EEPROM are integrated into one embedded system.

Individually calibrated and tested, **HCSD** performs $\pm 2\%$ from 20% to 80%RH ($\pm 3\%$ over entire humidity range), and yet, is simple and ready to use without further calibration or temperature compensation.

Designed and manufactured by industry leading humidity and temperature sensing technology of **SAMYOUNG S&C** – field proven in HVAC and Auto industry for over 10 years, **HCSD** offers another smart sensing solution for excellent reliability, high accuracy, and cost effective sensing applications.

Application

Energy Saving HVAC Control

Air Conditioning, Refrigeration, IAQ monitoring, Vent Fans, Home Appliances, Humi/Dehumidifiers

Process Control & Instrumentations

Medical Instruments, Handheld Devices, Weather Stations, Food Processing, Printers, RFIDs ...

Automobile & Transportation

Cabin Climate Control, Defogging Control
Condensing Preventive Device ...

Mass Quantity Application

OEM custom specification available

Sensor Performance

Relative Humidity (%RH)

Resolution	14 bit (0.01%RH)
Accuracy ¹	±2.0 %RH (20~80%RH)*
Repeatability	±0.2 %RH
Hysteresis	±2.0 %RH
Linearity	<2.0 %RH
Response time ²	Max 7.0 sec (τ 63%)
Temp Coefficient	0.05%RH (at 10~60°C, 10~90%RH)
Operating range	0 ~ 100 %RH (Non-Condensing)
Long term drift	<0.5 %RH/yr (Normal condition)

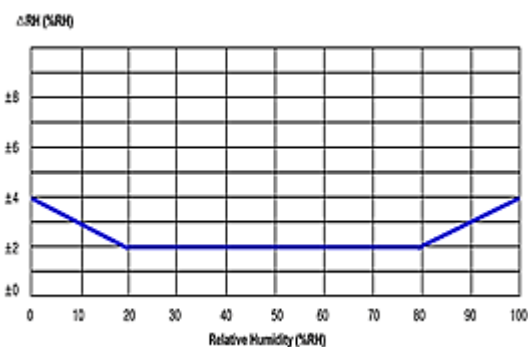
* Custom Accuracy Tolerance Available

Temperature(°C)

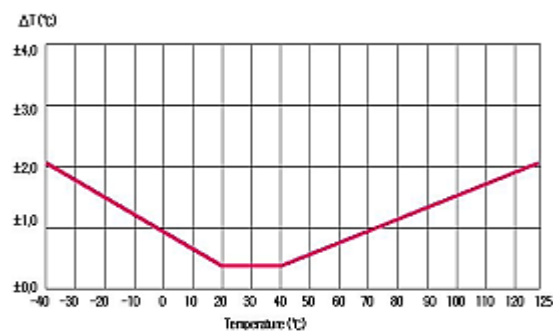
Resolution	14 bit (0.01°C)
Accuracy ³	±0.3 °C
Repeatability	±0.1 °C
Response time ⁴	5.0 sec (τ 63%)
Operating range	- 40 ~ 125 °C
Long term drift	<0.05 °C/yr (Normal condition)

1. Accuracies measured at 25°C, 5.0V.
2. Measured at 25°C, 1m/sec airflow for achieving 63% of step from 33%RH to 90%RH
3. Accuracies measured at 25°C, 5.0V.
4. Min 5.0 sec, Max 20 sec

Typical %RH Accuracy at 25°C

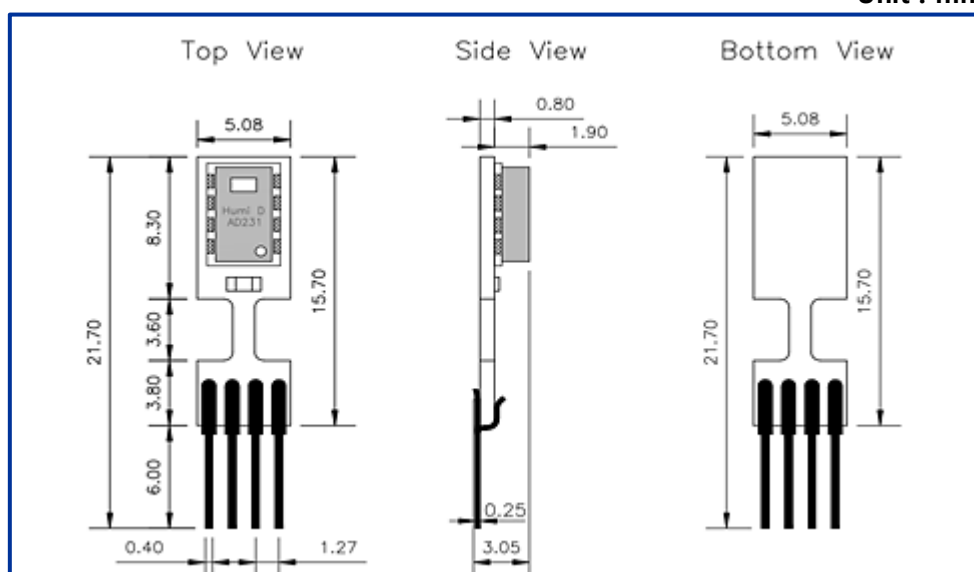


Typical Temperature Accuracy



Dimensions

Unit : mm



Electrical Specification

Supply Voltage ^{*1}	min 2.3V ~ max 5.5V
Supply Current (I _{DD})	750 μ A (typical)
Sleep Current (I _{sleep}) ^{*2}	0.6 μ A (typical)

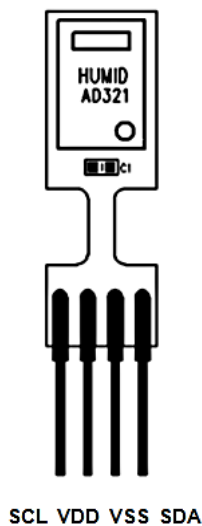
*1. Lower than 2.7V supply voltage may increase the accuracy tolerance of the Temperature.

*2. At -40~85°C

Environmental

Operating Temperature	-40 ~ 125°C
Operating Humidity	0~100%RH (non condensing)

Pin Connection



SCL VDD VSS SDA

* Chip on Board is 100nF capacitor for Vcore. (Figure. 4)

Absolute Maximum Rating

Parameter	Min	Max
Supply Voltage(V _{DD})	-0.3V	6.0V
Storage Temp	-55°C	150°C
Junction Temp	-55°C	150°C

Soldering Information

Standard or IR Solder Reflow.

IPC/JEDEC standard

T_P ≤ 250°C, t_P < 10sec, T_L < 220°C, t_L < 60sec.

T_P(Peak Temperature), t_P(Peak Time),

T_L(Critical Zone Temperature),

t_L(Critical Zone Time)

Ramp-up/down speed < 6°C/sec

Package Contents

HCSD series consists of a HumiChip and a V-core capacitor soldered on the top of a FR4 substrate. Lead Pins are made of Cu, Sn, P alloy and all parts are fully RoHS and REACH compliant.

Shipping

Tray	: 100ea
Inner Box	: 500ea
Out Box	: 2,500ea

Ordering Information

Output		Accuracy	VDD	MODE	Ordering P/N
D	I²C	±2%RH	3.3	Update	HCSD- 3V-U2
		±3%RH	5.0		HCSD- 5V-U3
		±2%RH	3.3	Sleep	HCSD- 3V-S2
		±3%RH	5.0		HCSD- 5V-S3

◆ For more detail **Part Numbers**,
Please refer to the [Table 14](#)

➤ **Standard Digital (I²C) Output :**

Factory setting is **Sleep Mode** and calibrated at **3.3V**

Application Guide

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Humidity and Temperature Conversion Formula

✦ **Demo Kit (EVB01-HCP)** is available.

A reference program source code can be downloaded from www.samyongsnc.com

1. General Information

1.1 Preliminary Consideration

To maximize the performance of **HCS**D, it is important to plan an appropriate location of the sensor at the design stage. Airflow and proper exposure to ambient air must be secured for **HCS**D to ensure expected performance. Airflow holes must

NOT be blocked. Any heat generating parts near **HCS**D will distort the proper measurement of relative humidity and temperature reading, and either should be avoided or measures should be taken to prevent heat transfer.

1.2 Operating Conditions

HCSD's maximum and recommended normal operating condition is shown below (**Figure 1**). Within the Normal Range **HCS**D performs stably. Prolonged exposures to conditions outside normal range, especially at humidity over 90%RH, may temporarily offset the RH signal up to $\pm 3\%$ RH. When return to Normal Range, it will gradually recover back to the calibration state.

Re-Conditioning Procedure in **Section 1.6** will help reduce this recovery time. Long term exposure to extreme conditions may also accelerate aging of the sensor.

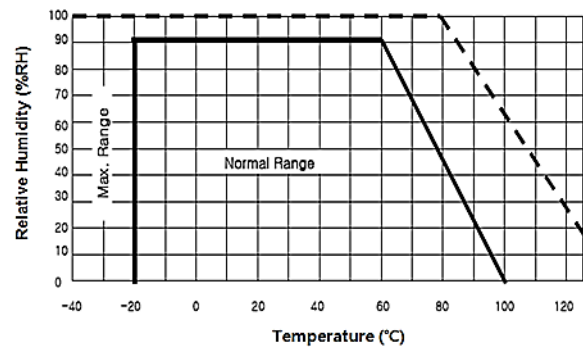


Figure 1 Operating Conditions

1.3 Heating

Though within the accuracy tolerance, Self-Heating in the sensor IC may affect accurate measurement of temperature and %RH. The measurement error from Self Heating can be reduced by keeping 'Active State' to the minimum, and by regulating the operating voltage within $3.3 \pm 0.5V$ or $5.0 \pm 0.5V$. In order to minimize the self heating, sampling time should be more than 180ms (200ms recommended).

Other heat sources such as power electronics, microcontrollers, and display near the sensor may affect the accurate measurement. The location of sensor near such heat sources should be avoided by maintaining distance or thermal buffer. Thin metal pattern, or even better, 'milling slits' around the sensor also may help reduce the error.

1.4 Storage and Handling Information

HCSD contains polymer based capacitive humidity sensor sensitive to environment, and should NOT be handled as an ordinary electronic component.

Chemical vapors at high concentration may interface with the polymer layers, and coupled with long exposure time, may cause a shift in both offset and sensitivity of the sensor.

Despite the sensor endures the extreme conditions of $-55^{\circ}C \sim 150^{\circ}C$, $0\% \sim 100\%RH$ (non condensing), long term exposure in such environment may also

offset the sensor reading. Hence, once the package is opened, it is recommended to store in clean environment of temperature at $5^{\circ}C \sim 55^{\circ}C$ and humidity at $10\% \sim 70\%RH$.

HCSD is protected of ESD up to 4000V and Latchup of $\pm 100mA$ or (up to +8V / down to -4V) relative to VSS/VSSA, and also packed in ESD protected shipping material. Normal ESD precaution is required when handling in assembly process.

1.5 Manual soldering

Contact time must be limited to 5 seconds at up to $350^{\circ}C$

1.6 Material Contents

HCSD consists of sensor cell and IC (polymer / glass & silicon substrate). The sensor housing consists of a PPS (Poly Phenylene Sulfide) cap with epoxy glob top on a standard FR4 substrate. Pins are made of a Cu/P/Sn alloy coated with Ni and Au.

The device is free of Pb, Cd and Hg.

RoHS compliant / REACH report available

1.7 Traceability Information

HCSD is laser marked with product type and lot identification. Further information about individual sensor is electronically stored on the chip. The first line denotes the sensor type: **HUMI-D** for **I²C** output.

Lot identification is printed on the second line with 5 digit alphanumeric code.

An electronic identification code stored on the chip can be decoded by **SAMYOUNG S&C** only and allows for tracking on batch level through production, calibration and testing.

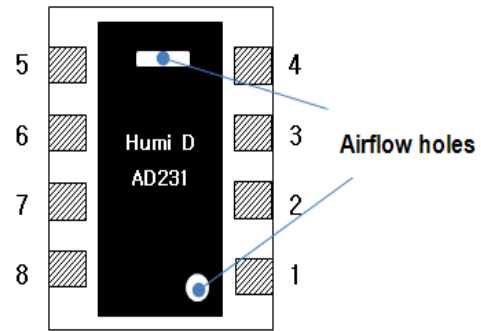


Figure 2 Laser Marking

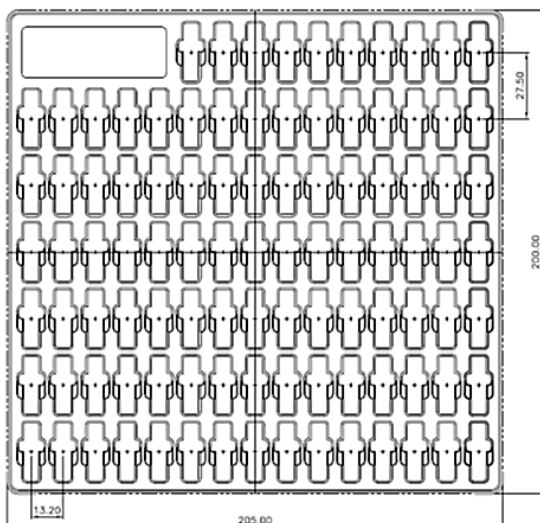
1.8 Shipping Package

HCSD is provided in a tray shipment packaging, sealed into antistatic ESD trays. The drawing of the

packaging tapes with sensor orientation and packing box dimensions are shown in **Drawing 2**

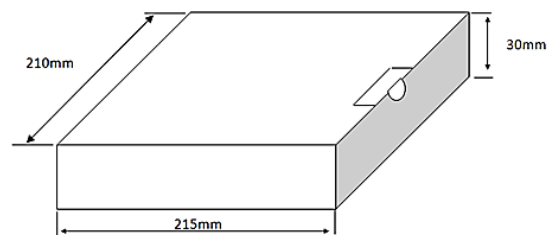
Drawing 1

Packing Tray : 100ea
Dimension : 205 x 200 mm

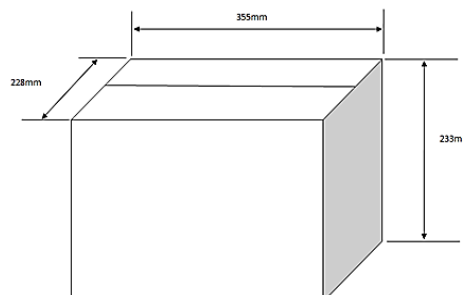


Drawing 2 Packing (Box)

Inbox : 500ea (5+1Tray(For Cover) X 100ea
Dimension : 215 x 210 x 30mm



Outbox : 2,500ea (5 X Inbox 500)
Dimension : 355 x 228 x 233mm



2. Interface Specification

2.1 Digital Output (I²C Interface)



Pin-No	Name	Description
1	SCL	I ² C clock
2	VDD	Supply voltage (2.3~5.5V)
3	VSS	Ground
4	SDA	I ² C data

2.1.1 Power Pads (2:VDD)

HCSD is capable of operating on wide range of power supply voltage from 2.3V to 5.5V. Recommended supply voltage is either $3.3\pm0.5V$ or

$5.0\pm0.5V$. Power supply should be connected to VDD (power supply pad 2). VDD and VSS (Ground pad 3) should be decoupled with a 220nF capacitor

2.1.2 Serial Clock & Data Pads (1:SCL, 4:SDA)

The sensor's data is transferred in and out through the SDA pad while the communication between **HCSD** and microcontroller (MCU) is synchronized through the SCL pad.

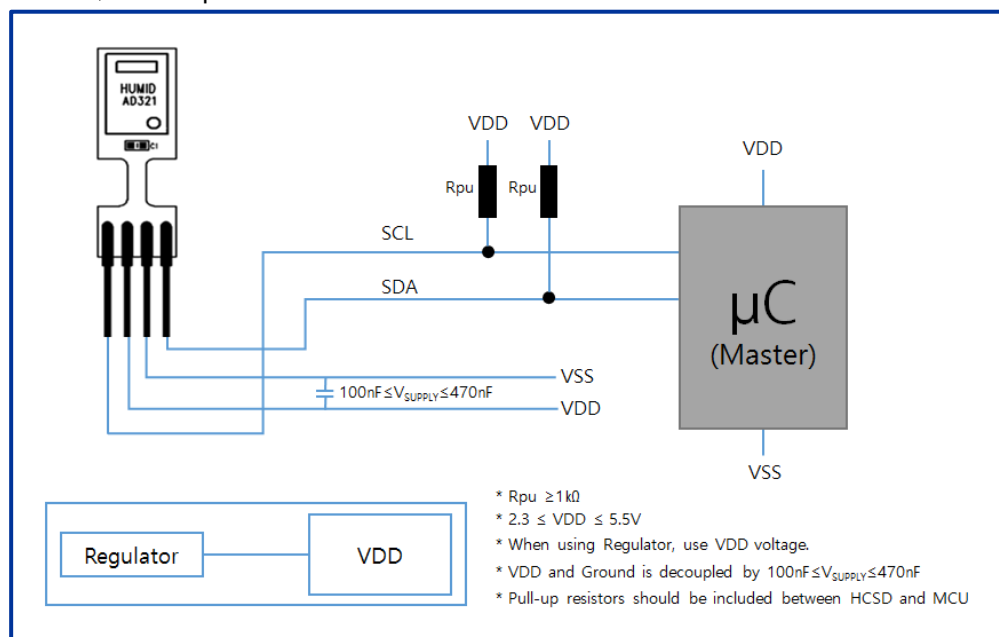
HCSD has an internal temperature compensated oscillator that provide time base for all operation, and uses an I²C-compatible communication protocol with support for 20kHz to 400kHz bit rates.

External pull-up resistors are required to pull the drive signal high, that can be included in I/O circuits of microcontroller. (see [Figure 3](#))

Further information about timing and communication between the sensor and microcontroller is explained in **Section 4**. Communicating with **HCSD**.

Figure 3 Typical Application Circuit (I²C)

- ▶ VDD and Ground is decoupled by a (100~470nF) capacitor.
- ▶ Pull-up resistor should be included between **HCSD** and MCU.



3. Electrical Specification

3.1 Absolute Maximum Rating

[Table 1](#) shows the Absolute Maximum Ratings for HCSD. Exposure to these extreme condition for

extended period may deteriorate the sensor performance and accelerate aging. Functional operation is not implied at these conditions.

3.2 Electrical Specification and Recommended Operating Conditions

The operating conditions recommended for HCSD is given in [Table 2](#) and the electrical specification is shown in [Table 3](#).

3.3 Output Pad Drive Strength

Output pad drive strength at different supply voltages and operating temperatures are shown in [Table 4](#) and [Table 5](#).

3.4 ESD/Latch-Up-Protection

All external module pins have an ESD protection of up to 4000V and a latch-up protection of $\pm 100\text{mA}$ or (up to +8V / down to -4V) relative to VSS/VSSA. The internal module pin V_{CORE} has ESD protection of up

to 2000V. The ESD test follows the Human Body Model with 1.5k Ω /100pF based on MIL 883, Method 3015.7.

Table 1 Absolute Maximum Rating

PARAMETER	SYMBOL	MIN	MAX	Unit
Supply Voltage (V _{DD})	V _{DD}	-0.3	6.0	V
Storage Voltage at I/O pads	T _{IO}	-0.3	V _{DD} +0.3	V
Storage Temperature Range	T _{STOR}	-55	150	°C
Junction Temperature	T _J	-55	150	°C

Table 2 Recommended Operating Conditions

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
Supply Voltage	V _{SUPPLY}	2.3		5.5	V
Ambient Temperature Range	T _{AMB}	-40		125	°C
External Capacitance between V _{DD} pin and Gnd	C _{V_{SUPPLY}}	100	220	470	nF
External Capacitance between V _{CORE} and Gnd	C _{V_{CORE}}		100		nF
Pull-up on SDA and SCL	R _{PU}	1	2.2		k Ω

Table 3 Electrical Characteristics Specifications

PARAMETER	SYMBOL	CONDITIONS	MIN.	TYP.	MAX.	UNITS
SUPPLY						
Supply Current (varies with update rate and output mode)	I _{DD}	At maximum update rate		750	1100	μA
Sleep Mode Current	I _{sleep}	-40 to 85°C		0.6	1	μA
		-40 to 125°C		1	3	μA
DIGITAL I/O						
Voltage Output Level Low	V _{OL}			0	0.2	V _{SUPPLY}
Voltage Output Level High	V _{OH}		0.8	1		V _{SUPPLY}
Voltage Input Level Low	V _{IL}			0	0.2	V _{SUPPLY}
Voltage Input Level High	V _{IH}		0.8	1		V _{SUPPLY}
TOTAL SYSTEM						
Start-Up-Time Power-on (POR) to data ready	t _{STA}	At nominal frequency; fastest and slowest setting	4.25		55	ms
Update Rate (Update Mode)	t _{RESP_UP}	Fastest and slowest settings	0.70		165	ms
Response Time (Sleep Mode)	t _{RESP_SL}	Fastest and slowest settings	1.25		45	ms

Table 4 Output High Drive Strength

Output High Drive Strength (mA)						
V _{SUPPLY} (V)	-40°C		25°C		125°C	
	Min	Typ	Min	Typ	Min	Typ
2.3	3.8	6.2	3.3	5.1	2.8	4.2
2.7	7.2	10.5	5.9	8.4	4.7	6.6
3.3	12.1	16.6	9.6	12.9	7.4	10.0
5.5	20.0	20.0	20.0	20.0	20.0	20.0

Table 5 Output Low Drive Strength

Output Low Drive Strength (mA)						
V _{SUPPLY} (V)	-40°C		25°C		125°C	
	Min	Typ	Min	Typ	Min	Typ
2.3	10.8	16.0	8.8	12.6	6.9	9.5
2.7	20.0	20.0	16.0	20.0	11.7	14.9
3.3	20.0	20.0	20.0	20.0	18.2	20.0
5.5	20.0	20.0	20.0	20.0	20.0	20.0

4. Communicating With HCSD

4.1 Power-On Sequence

On system power-on reset (POR), the **HCSD** wakes as an I²C device regardless of the output protocol programmed in EEPROM. After power-on reset, it enters the command window. It then waits for a Start_CM command for 10ms if Fast Startup bit is not set in EEPROM (Factory Setting) or for 3ms if fast startup bit is set in EEPROM (see **Figure 4**). If the **HCSD** receives the Start_CM command during the command window, it enters and remains in Command Mode.

Command Mode is primarily used for initializing **HCSD**.

If during the power-on sequence, the command window expires without receiving a Start_CM or if the part receives a Start_NOM command in Command Mode, the device will immediately assume its programmed output mode and will perform one complete measurement cycle.

Figure 4 General Operation

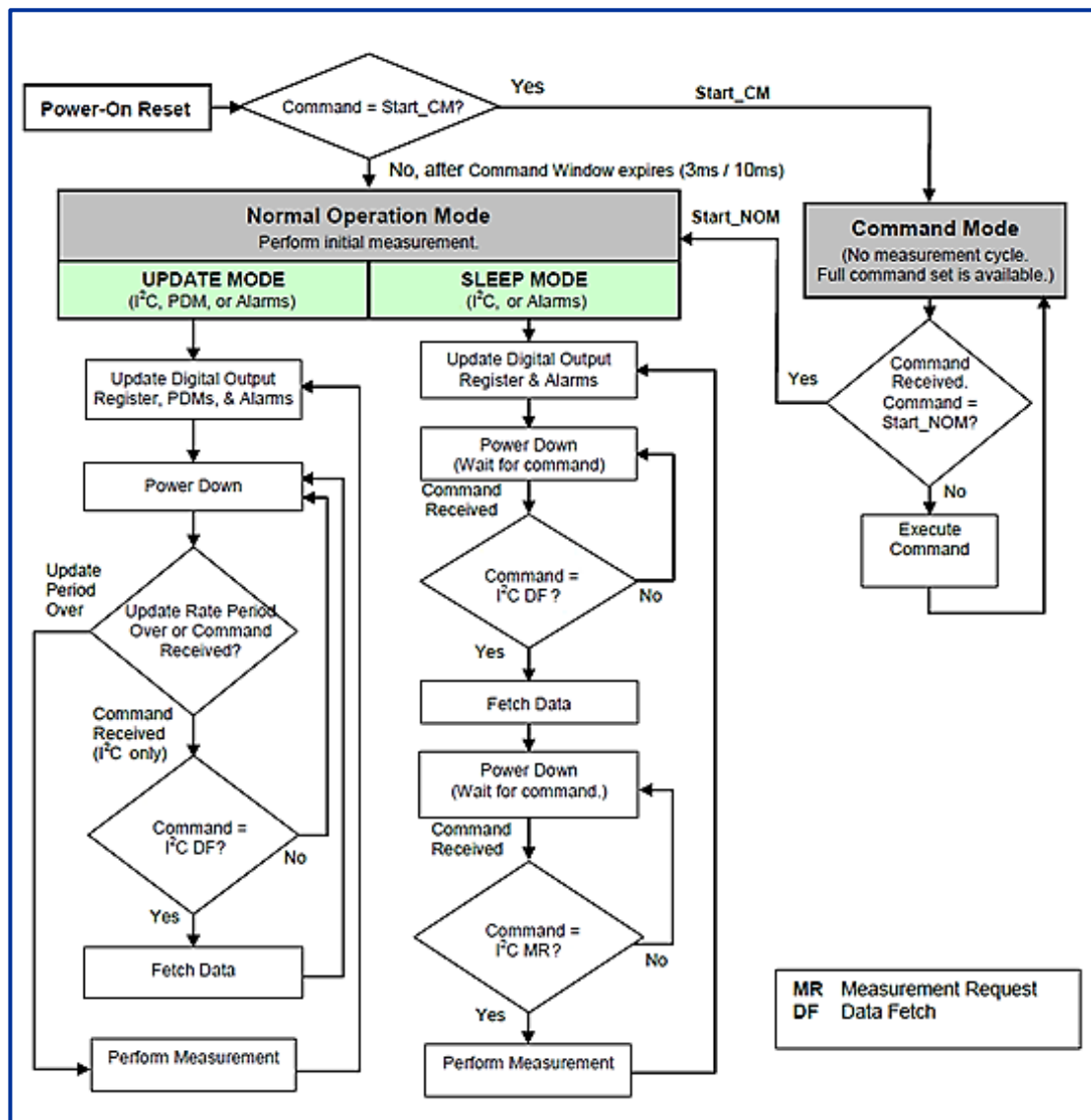


Figure 5 Power-On Sequence with Fast Start-up Bit Set in EEPROM

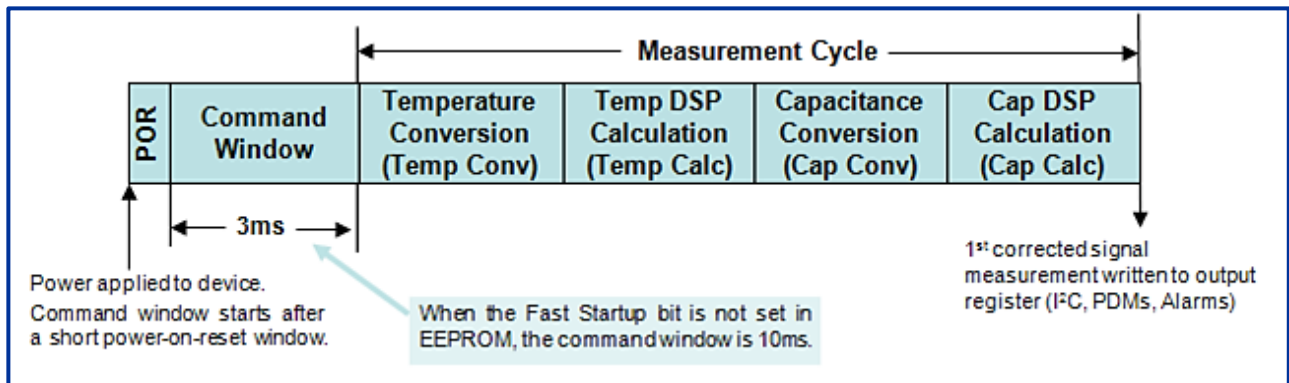
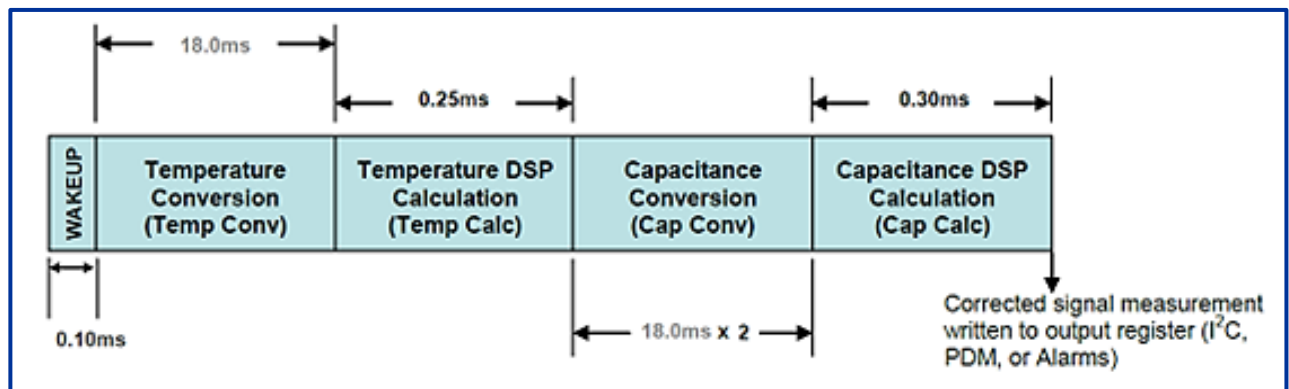


Figure 6 Measurement Cycle Timing



4.2 I²C Features and Timing

HCSD uses I²C-compatible communication protocol with support for 20kHz and 400kHz bit rates. The I²C slave address (0x00 to 0x7F) is selected by the Device_ID bits in the Cust_Config EEPROM word (see [Table 13](#) for bit assignments).

See [Figure 7](#) for I²C Timing Diagram and [Table 6](#) for definitions of the parameters shown in the diagram.

Note: Detailed Timing Chart and Reference Programming Code are available upon request.

Figure 7 I²C Timing Diagram

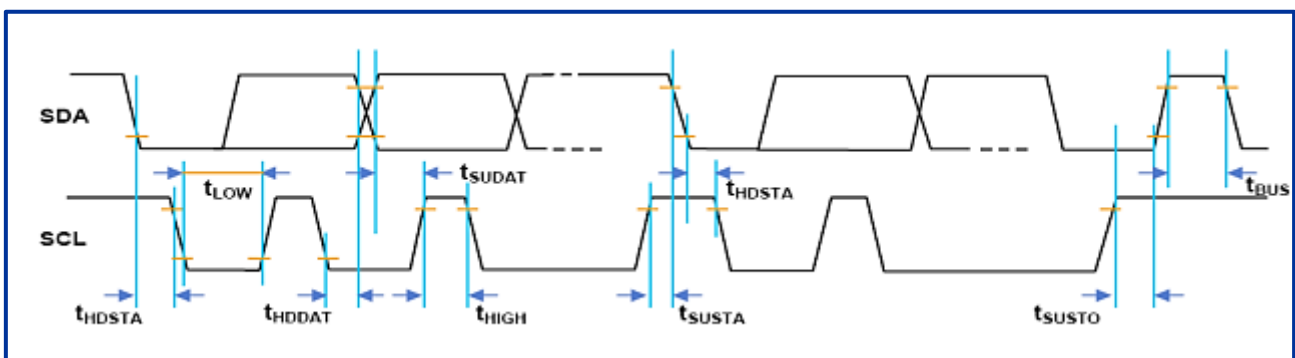


Table 6 I²C Parameters

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
SCL clock frequency	f _{SCL}	20		400	kHz
Start condition hold time relative to SCL edge	t _{DHSTA}	0.1			μs
Minimum SCL clock low width ¹	t _{LOW}	0.6			μs
Minimum SCL clock high width ¹	t _{HIGH}	0.6			μs
Start condition setup time relative to SCL edge	t _{SUSTA}	0.1			μs
Data hold time on SDA relative to SCL edge	t _{HDDAT}	0		0.5	μs
Data setup time on SDA relative to SCL edge	t _{SUDAT}	0.1			μs
Stop condition setup time on SCL	t _{SUSTO}	0.1			μs
Bus free time between stop condition and start condition	t _{BUS}	1			μs

1. Combined low and high widths must equal or exceed minimum SCL period.

4.3 Measurement Modes

The **HCS**D can be programmed to operate in either Sleep Mode or Update Mode. The measurement mode is selected with the Measurement_Mode bit in

the **HCS**D Config Register word. In Sleep Mode, the part waits for commands from the master before taking measurements (see **section 4.3.2**).

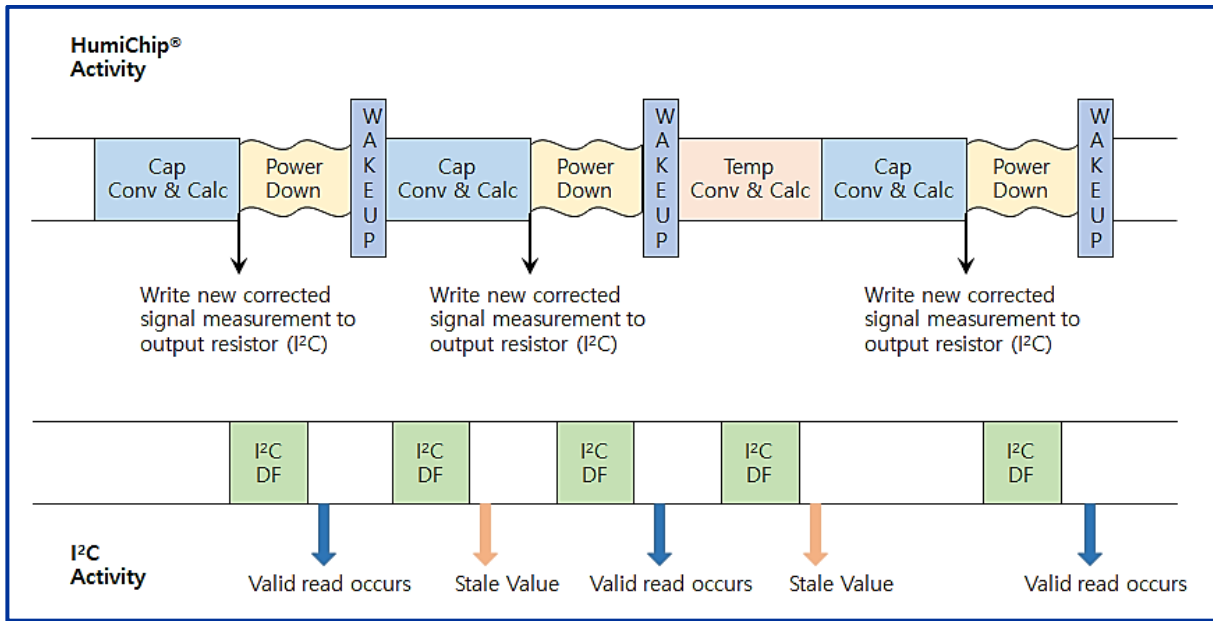
4.3.1 Data Fetch in Update Mode

In Update Mode, I²C is used to fetch data from the digital output register using a Data Fetch (DF) command.

Detecting when data is ready to be fetched can be handled by polling. The status bits of a DF tell

whether the data is valid or stale (see [Table 7](#) regarding the status bits). As shown in [Figure 8](#) after a measurement cycle is complete, valid data can be fetched. If the next data fetch is performed too early, the data will be the same as the previous fetch with stale status bits.

Figure 8 I²C Data Fetching in Update Mode



4.3.2 Data Fetch in Sleep Mode

In Sleep Mode, the **HCSD Core** will only perform conversions when **HCSD** receives a Measurement Request command (MR); otherwise, the **HCSD** is always powered down. Measurement Request commands can only be sent using I²C.

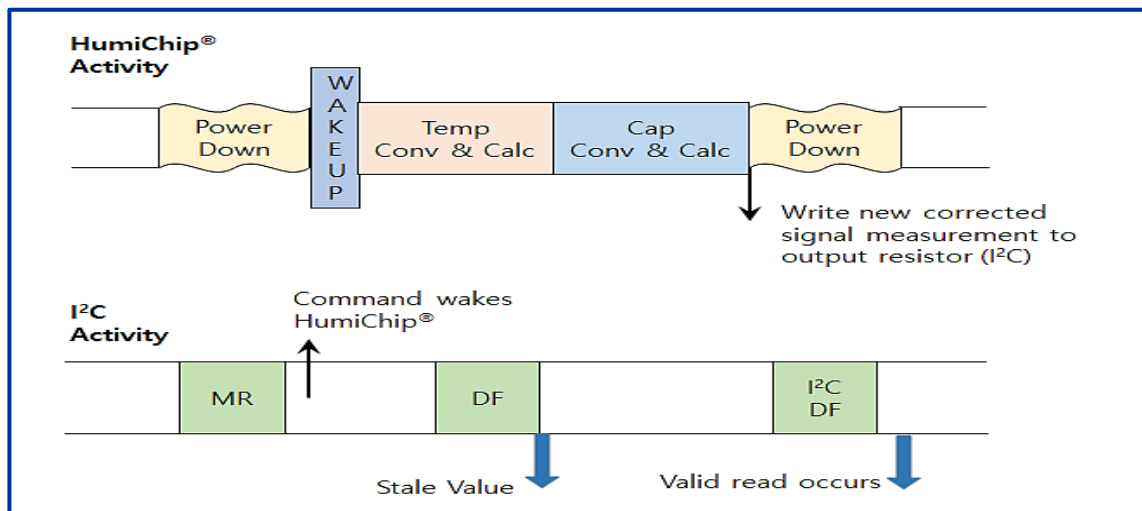
Note: Sleep Mode power consumption is significantly lower than Update Mode power consumption (see [Table 3](#) for exact values).

Figure 9 shows the measurement and communication sequence for Sleep Mode. The master sends an MR command to wake the **HCSD**

from power down. After **HCSD** wakes up, a measurement cycle is performed consisting of both a temperature and a capacitance conversion followed by the **HCSD Core** correction calculations.

An I²C data fetch (DF) is performed during the power-down period to fetch the data from the output register. In I²C the user can send another MR to start a new measurement cycle without fetching the previous data. After the data has been fetched, the **HCSD** remains powered down until the master sends an MR command.

Figure 9 I²C Data Fetching in Sleep Mode



In Sleep Mode, I²C are used to request a measurement with a MR command and to fetch data from the digital output register using a Data Fetch (DF) command (see **Section 4.7** for details on the MR command).

The status bits of a DF tell whether the data is valid or stale (see **Section 4.4** regarding the status bits). As shown in **Figure 9** after a measurement cycle is complete, valid data can be fetched. If the next data fetch is performed too early, the data will be the same as the previous fetch with stale status bits.

Note : There is an 18 μ s period when the measurement cycle completes that must have the bus unlocked. So wait for the measurement cycle to complete prior to reading the I²C bus. This can be determined using either the ready pin or by waiting a fixed amount of time as specified in **Figure 6**. Or use an I²C frequency of 60kHz or greater.

4.4 Status Bits

Status bits (the two MSBs of the fetched high data byte, see **Table 7**) are provided in I²C. The status

bits are used to indicate the current state of the fetched data.

Table 7 Status Bits

Status Bits (I ² C)	Definition
00 _B	Valid data : Data that has not been fetched since the last measurement cycle.
01 _B	Stale data : Data that has already been fetched since the last measurement cycle.
10 _B	Command Mode : The HCSD is in Command Mode.
11 _B	Not used

Note : '%' symbol in Checksum Formula means mod operation.

4.5 I²C Commands

As detailed in **Table 8**, there are two types of commands which allow the user to interface with

the HCSD in the I²C.

Table 8 I²C Command Bits

Type	Description	Communication Supported	Reference Sections
Data Fetch (DF)	Used to fetch data in any digital mode	I ² C	Section 4.6
Measurement Request (MR)	Used to start measurements in Sleep Mode	I ² C	Section 4.7

4.6 Data Fetch (DF)

The Data Fetch (DF) command is used to fetch data in any digital output mode.

An I²C Data Fetch command starts with the 7-bit slave address and the 8th bit = 1 (READ).

The HCSD as the slave sends an acknowledgement (ACK) indicating success.

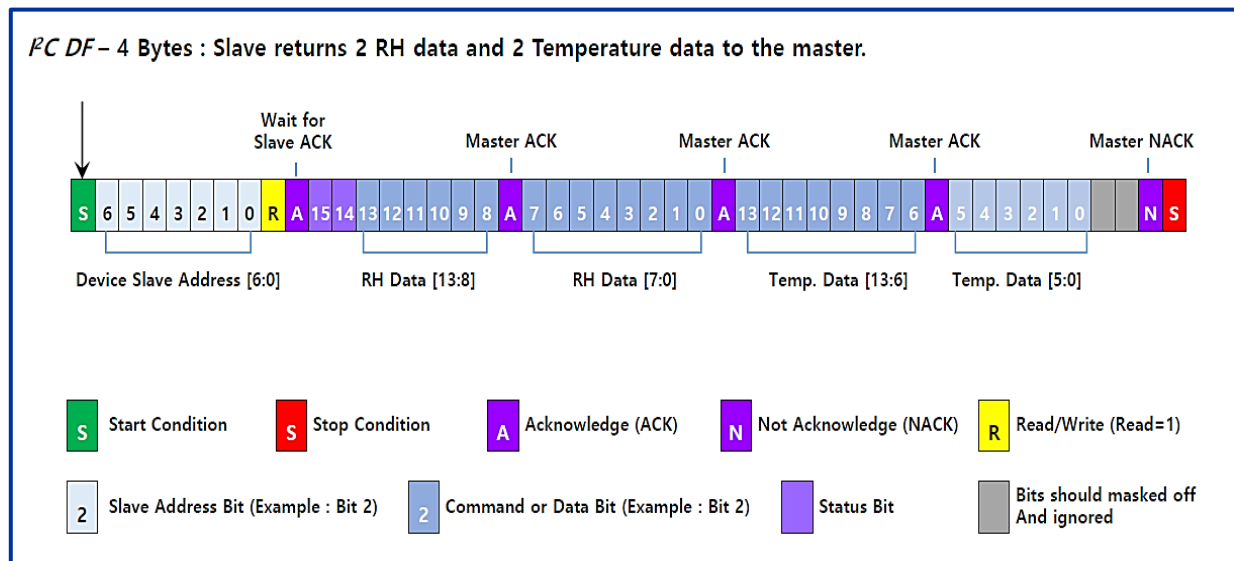
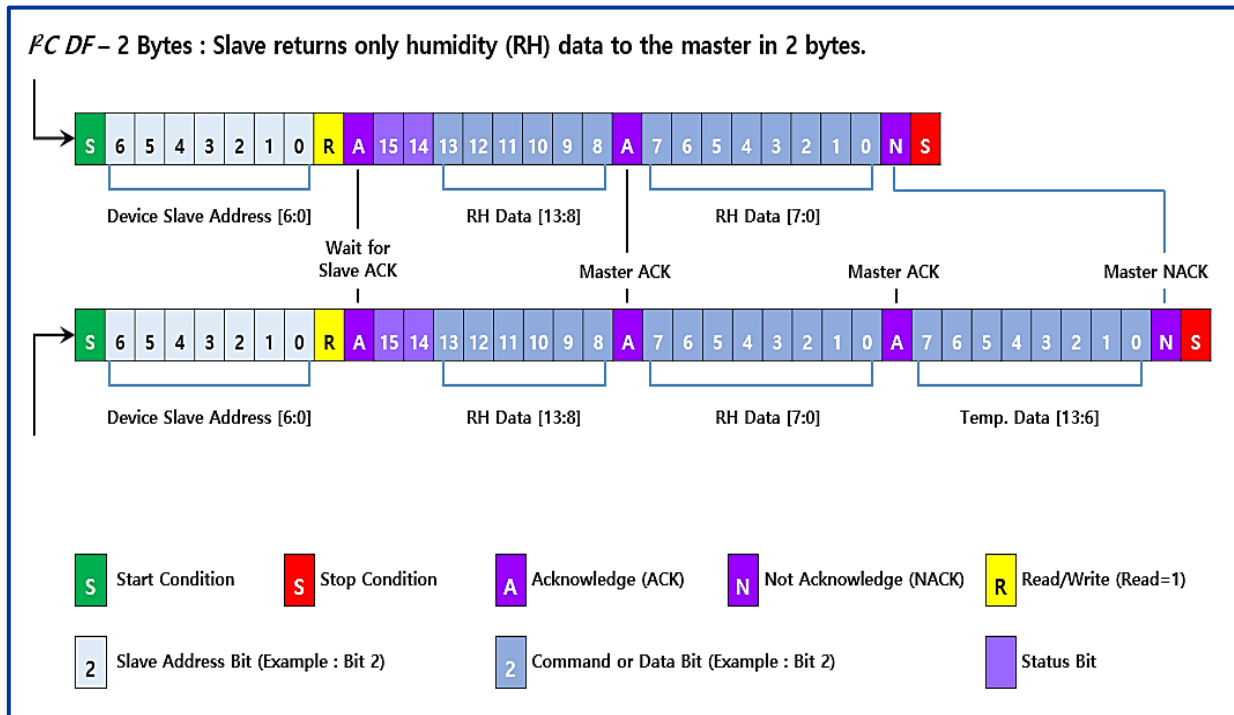
The number of data bytes returned by the HCSD is determined by when the master sends the NACK and stop condition. **Figure 10** shows examples of fetching two, three and four bytes respectively.

The full 14 bits of humidity data are fetched in the

first two bytes. The MSBs of the first byte are the status bits.

If temperature data is needed, additional temperature bytes can be fetched. In **Figure 10**, the three-byte data fetch returns 1 byte of temperature data (8-bit accuracy) after the humidity data. A fourth byte can be fetched where the six MSBs of the fetched byte are the six LSBs of a 14-bit temperature measurement. The last two bits of the fourth byte are undetermined and should be masked off in the application.

Figure 10 I²C Measurement Packet Reads



Humidity & Temperature Conversion Formula

Humidity Output (%RH)	$(RH_High [5:0] \times 256 + RH_Low [7:0]) / 2^{14} \times 100$
Temperature Output (°C)	$(Temp_High [7:0] \times 64 + Temp_Low [7:2] / 4) / 2^{14} \times 165 - 40$

4.7 Measurement Request (MR)

A measurement request (MR) is a Sleep-Mode-only command sent by the master to wake up the **HCSD** and start a new measurement cycle in I²C.

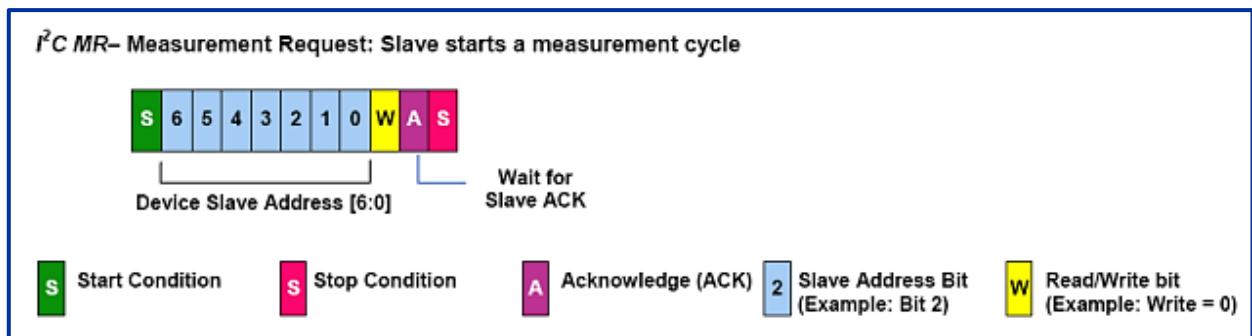
The I²C MR is used to wake up the device in Sleep Mode and start a complete measurement cycle starting with a temperature measurement, followed by a humidity measurement, and then the results can be fetched by master with I²C.

As shown in **Figure 11**, the communication contains only the slave address and the WRITE bit (0) sent by the master.

After the **HCSD** responds with the slave ACK, the master creates a stop condition.

Note: The I²C MR function can also be accomplished by sending “don’t care” data after the address instead of immediately sending a stop bit.

Figure 11 I²C Measurement Request

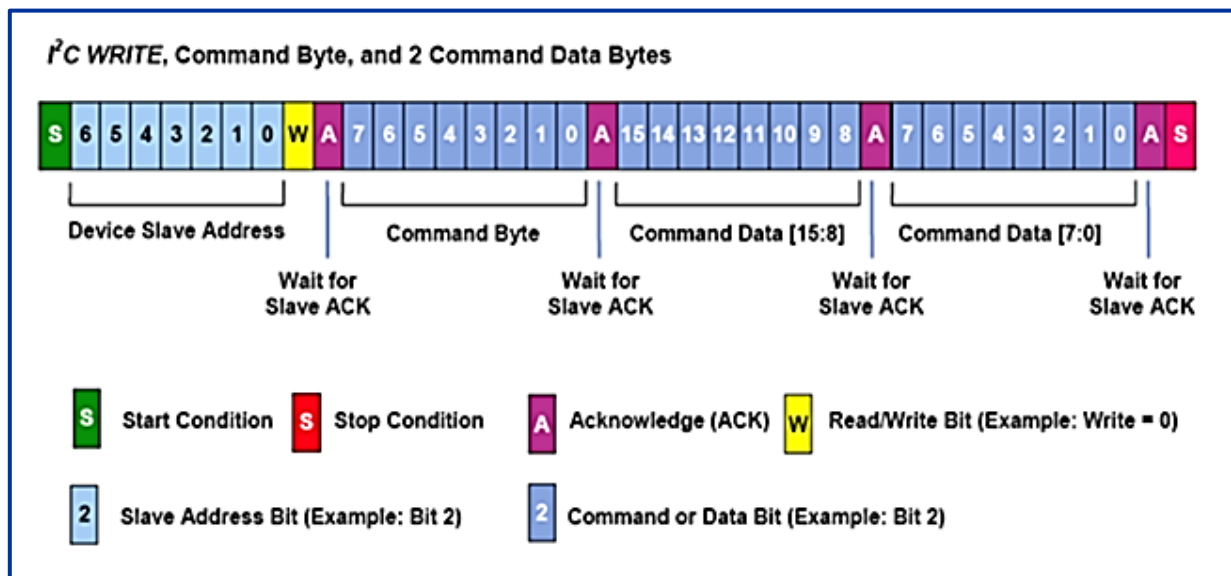


4.8 Command Mode

Command Mode commands are only supported for the I²C protocol. As shown in **Figure 12**, commands are 4-byte packets with the first byte being a 7-bit

slave address followed by 0 for write. The second byte is the command byte and the last two bytes form a 16-bit data field.

Figure 12 I²C Command Format



4.9 Command Encodings

Table 9 describes all the commands that are offered in Command Mode.

Note: Only the commands listed in **Table 9** are valid. Other encodings might cause unpredictable results. If data is not needed for the command, zeros must be supplied as data to complete the 4-byte packet.

Table 9 Command List and Encodings

Command Byte 8 Command Bits (Hex)	Third and Fourth Bytes 16 Data Bits (Hex)	Description	Response Time
0x1C to 0x1F	0x0000	EEPROM Read of addresses 0x1C to 0x1F After this command has been sent and executed, a data fetch must be performed.	100 μ s
0x5C to 0x5F	0xYYYY (Y = data)	Write to EEPROM addresses 0x1C to 0x1F The 2 bytes of data sent will be written to the address specified in the 6 LSBs of the command byte.	12ms
0x80	0x0000	Start_NOM Ends Command Mode and transitions to Normal Operation Mode.	
0xA0	0x0000	Start_CM Start Command Mode: used to enter the command interpreting mode. Start_CM is only valid during the power-on command window.	100 μ s

4.10 Command Response and Data Retch

After a command has been sent and the execution time defined in **Table 9** has expired, an I²C Data Fetch (DF) can be performed to fetch the response. As shown in **Figure 13**, after the slave address has been sent, the first byte fetched is the response byte.

The upper two status bits will always be 10B to represent Command Mode. The lower two bits are the response bits. **Table 10** describes the different responses that can be fetched. To determine if a command has finished executing, poll the part until a Busy response is no longer received. The middle four bits of the response byte are command diagnostic bits where each bit represents a different diagnostic (see **Table 11**).

Note: Regardless of what the response bits are, one or more of the diagnostic bits may be set indicating an error occurred during the execution of the command.

Note: Only one command can be executed at a time. After a command is sent another command must not be sent until the execution time of the first command defined in **Table 9** has expired.

For all commands except EEPROM Read, the data fetch should be terminated after the response byte is read. If the command was an EEPROM Read, then the user will fetch two more bytes as shown in **Figure 13**, example 3. If a Corrected EEPROM Error diagnostic was flagged after an EEPROM read, the user has the option to write this data back to attempt to fix the error.

Then a data fetch can be performed to get the response and data (see **Figure 13**).

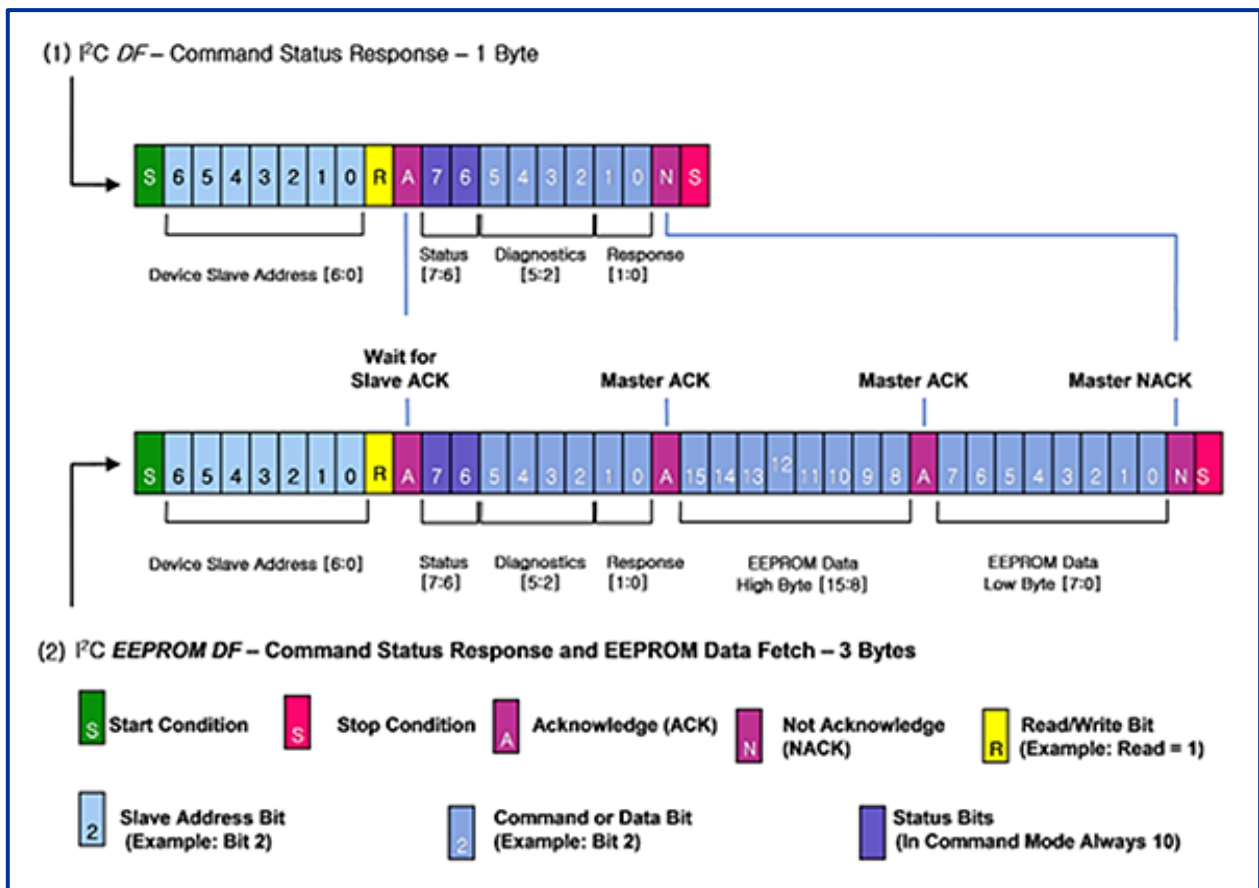
Table 10 Response Bits

Encoding	Name	Description
00 _B	Busy	The command is busy executing.
01 _B	Positive Acknowledge	The command executed successfully.
10 _B	Negative Acknowledge	The command was not recognized or an EEPROM write was attempted while the EEPROM was locked.

Table 11 Command Diagnostic Bits

Bit Position	Name	Description
2	Corrected EEPROM Error	A corrected EEPROM error occurred in execution of the last command.
3	Uncorrectable EEPROM Error	An uncorrectable EEPROM error occurred in execution of the last command.
4	RAM Parity Error	A RAM parity error occurred during a microcontroller instruction in the execution of the last command.
5	Configuration Error	An EEPROM or RAM parity error occurred in the initial loading of the configuration registers.

Figure 13 Command Mode Data Fetch



4.11 EEPROM

The EEPROM array contains the calibration coefficients for gain and offset, etc., and the configuration bits for the analog front end, output modes, measurement modes, etc. The **HCS**D EEPROM is arranged as 4 16-bit words (see [Table 12](#)).

See **Section 4.9** Command Mode for instructions on reading and writing to the EEPROM in Command Mode via the I²C interface. When programming the EEPROM, an internal charge pump voltage is used; therefore a high voltage supply is not needed.

Table 12 EEPROM Word Assignments

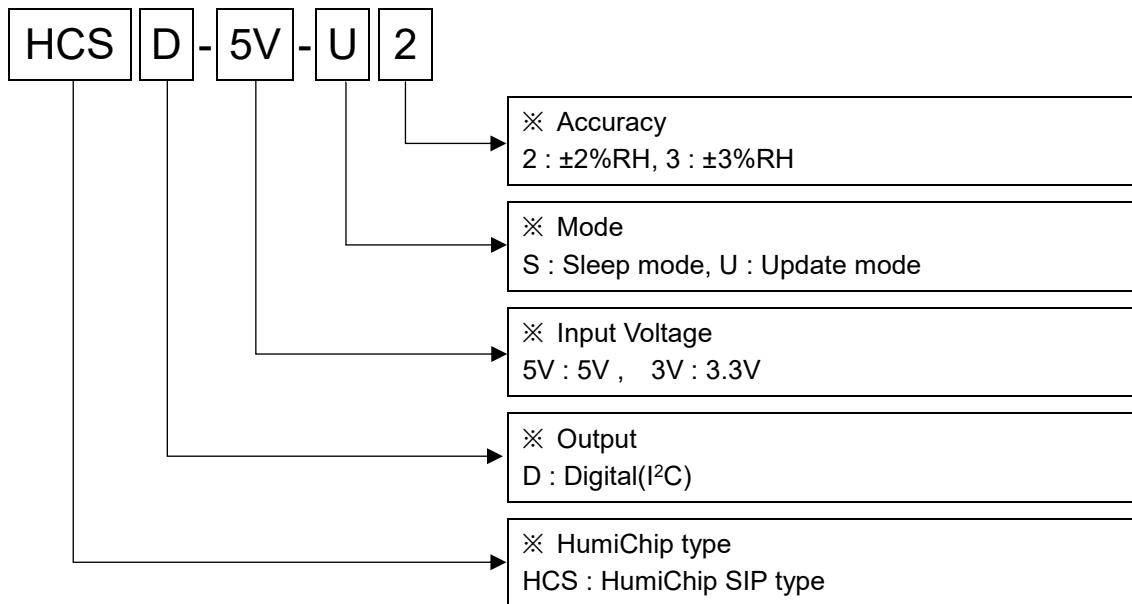
EEPROM Word	Bit Range	IC Default	Name	Description and Notes
1C _{HEX}	15:0	0x0028	Cust_Config	Customer Configuration (see Table 13)
1D _{HEX}	15:0	0x0000	Reserved	Reserved Word: Do Not Change ; must leave at factory settings
1E _{HEX}	15:0	0x0000	Cust_ID2	Customer ID byte 2: For use by customer
1F _{HEX}	15:0	0x0000	Cust_ID3	Customer ID byte 3: For use by customer

Table 13 Cust_Config Bit Assignments

Bit Range	IC Default	Name	Description and Notes
6:0	0101000	Device_ID	I ² C slave address
12:0	000000	Reserved	Do Not Change – must leave at factory settings
13	0	Fast_Startup	Sets the Command Window length: 0 = 10 ms Command Window 1 = 3 ms Command Window
15:14	00	Reserved	Do Not Change – must leave at factory settings

Table 14

HCSD Series Part Number List



Items	Series	Output	Operating Voltage(V)	mode	Accuracy(%RH)	Part Number
Humi D	HCS (SIP)	I ² C (Digital)	5	Update	±2	HCSD-5V-U2
			5	Update	±3	HCSD-5V-U3
			3.3	Update	±2	HCSD-3V-U2
			3.3	Update	±3	HCSD-3V-U3
			5	Sleep	±2	HCSD-5V-S2
			5	Sleep	±3	HCSD-5V-S3
			3.3	Sleep	±2	HCSD-3V-S2
			3.3	Sleep	±3	HCSD-3V-S3

5. Revision History

Date	Version	Page(s)	Changes
16 SEP 2015	1.0		First Release
27 APR 2017	4.0	ALL	Font & drawing modified



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